A Game-Theoretic Analysis of MEV in Decentralised Exchanges

Progress Report 1

Name	Roll Number
Chaitanya Garg	22B0979
Chinmay Khokar	22B1027
Sanoj S Vijendra	22B0916

February 22, 2026

Abstract

The report presents a comprehensive literature review on Maximal Extractable Value (MEV) and its game theoretic dimensions in decentralised exchange (DEX) ecosystems, followed by a rigorous formal model of the strategic game among users, miners/validators, and DEX protocols. We survey foundational and recent works on frontrunning, Priority Gas Auctions (PGAs), MEV driven consensus instability, and batch-auction-based mitigations. We define the players, strategy spaces, payoff functions and equilibrium concepts that will underpin our simulation framework later. The theoretical model is cast as a three player Stackelberg game under continuous trading, with an extension to batch auction clearing, enabling a comparative welfare analysis across protocol designs.

Contents

1 Introduction **3**
1.1 Progress Made so Far . . . 3

2 Literature Review **3**
2.1 MEV: Origins and Measurement . . . 3
2.1.1 Pure Revenue Opportunities . . . 3
2.1.2 Priority Gas Auctions (PGAs) . . . 4
2.1.3 MEV and Consensus Instability . . . 4
2.2 Fair Ordering and MEV Mitigation . . . 4
2.2.1 Randomised Sequencing . . . 4
2.2.2 Proposer-Builder Separation (PBS) . . . 5
2.3 Batch Auctions as a Market Design Response . . . 5
2.4 Related Protocols and Extensions . . . 5
2.5 Summary of Literature Gaps . . . 6

3 Formal Game Theoretic Model **6**
3.1 Overview and Modelling Philosophy . . . 6
3.2 State Space . . . 7
3.3 Players . . . 7
3.3.1 Player 1: The User Set . . . 7
3.3.2 Player 2: The Miner/Validator . . . 8
3.3.3 Player 3: The DEX Protocol . . . 8
3.4 Strategy Spaces . . . 8
3.4.1 User Strategy Space . . . 8
3.4.2 Miner Strategy Space . . . 9
3.4.3 Sandwich Attack: Formal Specification . . . 9
3.5 Payoff Functions . . . 10
3.5.1 User Payoff . . . 10
3.5.2 Miner Payoff . . . 10
3.5.3 Protocol Payoff / Social Welfare . . . 11
3.6 Information Structure . . . 11
3.7 Equilibrium Concepts . . . 11
3.7.1 Nash Equilibrium . . . 11
3.7.2 Stackelberg Equilibrium . . . 12
3.7.3 Repeated Game and Cooperation . . . 12
3.8 Priority Gas Auction Sub-Game . . . 12
3.9 The Batch Auction Extension . . . 13

4 Comparative Analysis: Continuous vs. Batch Clearing **13**
4.1 Equilibrium Outcomes Under Continuous Trading . . . 13
4.2 Equilibrium Outcomes Under Batch Auctions . . . 14
4.3 Welfare Comparison . . . 14

5 Discussion and Next Steps **15**
5.1 Key Theoretical Insights . . . 15
5.2 Plan for Phase 2 . . . 15

6 Conclusion **15**

1 Introduction

Decentralised exchanges (DEXs) have emerged as a cornerstone of the Decentralised Finance (DeFi) ecosystem enabling non-custodial, trustless token swapping through on-chain smart contracts. Unlike their centralised counterparts, DEXs expose the full transaction pipeline from submission to final settlement to a public peer-to-peer mempool, creating rich informational asymmetries that rational actors exploit for profit.

The most consequential of these exploits fall under the umbrella of *Maximal Extractable Value* (MEV): the surplus value that miners or validators can capture by selectively reordering, inserting, or censoring transactions within a block. MEV manifests in several forms, most notably *frontrunning*, *backrunning*, and *sandwich attacks*, each of which directly reduces user welfare while enriching block producers and sophisticated arbitrage bots.

1.1 Progress Made so Far

This first progress report covers:

- A thorough literature review (§2) covering MEV measurement, Priority Gas Auctions (PGAs), consensus-layer risks, and batch-auction alternatives.
- A complete formal game-theoretic model (§3), including player definitions, strategy spaces, payoff functions, information structures, and equilibrium concepts.
- A comparative discussion of continuous vs. batch-clearing equilibria (§4).

Note: Python simulation scripts are deferred to the second progress report due to unavoidable circumstances.

2 Literature Review

2.1 MEV: Origins and Measurement

The concept of Miner Extractable Value was formally introduced and empirically grounded by **Daian et al.** [1] in the landmark paper *Flash Boys 2.0: Frontrunning, Transaction Reordering, and Consensus Instability in Decentralised Exchanges*. Drawing on nine months of Ethereum mempool data collected from six geographically distributed nodes, the authors documented a thriving ecosystem of arbitrage bots exploiting price inefficiencies across multiple DEXs.

2.1.1 Pure Revenue Opportunities

Daian et al. identify a tractable sub class of MEV called *pure revenue opportunities*: atomic smart-contract transactions that execute multiple trades across one or more exchanges and yield positive profit *in every traded asset*, regardless of market movement. The all or nothing atomicity of Ethereum smart contracts is key, if any leg of the trade fails, the entire transaction reverts, eliminating execution risk.

The authors establish a lower bound of **over $6M USD** in pure revenue captured by bots on Ethereum across the observation period (July 2017–March 2019), with 1–10 ETH available for arbitrage on a typical day and occasional windfalls exceeding 100 ETH in a

single block (e.g., block 7,029,147, where a single arbitrage trade generated 101.6 ETH, more than 33× the block reward).

2.1.2 Priority Gas Auctions (PGAs)

Because pure revenue opportunities are broadcast publicly via the peer-to-peer gossip network, competing bots race to be the first to exploit them. This race takes the form of a **Priority Gas Auction (PGA)** where bots submit the same transaction repeatedly with increasing gas prices, bidding for priority inclusion in the next block.

Key empirical observations from Daian et al. include:

- PGAs are continuous-time, partial-information games with *all-pay* characteristics. Even losing bids consume gas fees.
- Bots converge over time to a **minimum bid-raise strategy** (12.5%, the Parity client's enforced minimum increment), consistent with a cooperative Nash equilibrium.
- Latency is a critical competitive dimension, mean inter-bid times fell from ∼0.8 s in mid-2018 to ∼0.1 s by early 2019.
- A **grim-trigger cooperative equilibrium** emerges organically on-chain, where players alternate minimum-increment bids and threaten to raise to the maximum if the other defects.

2.1.3 MEV and Consensus Instability

Beyond user welfare concerns, Daian et al. demonstrate that high MEV creates systemic risks at the *consensus layer* such as:

- **Undercutting attacks**: A miner observing a block with large Order Optimisation (OO) fees can profitably fork that block, withholding some fees to attract competing miners to build on the fork. The authors show this is a *present* threat because OO fees already dominate block rewards in some blocks.
- **Time-bandit attacks**: A miner can rewrite blockchain history to retroactively capture MEV locked in past blocks. With DEX volumes peaking at $1.5B/month and a 51% attack on Ethereum estimated at $56M/month (as of 2019), the economics of such attacks are plausible in high-MEV regimes.

2.2 Fair Ordering and MEV Mitigation

2.2.1 Randomised Sequencing

Alipanahloo, Zhang, and Awosika [2] propose *fair ordering policies* for Layer-2 rollups that randomise transaction sequencing to reduce the predictability that MEV extraction relies upon. They introduce the notion of *fairness granularity*: the minimum resolution at which a sequencer can distinguish and reorder transactions. Their key finding is that coarser granularity (i.e., batching transactions over wider time windows before ordering) reduces MEV opportunities but also increases latency.

While this work provides formal fairness guarantees and quantifies the MEV-latency trade-off, it does not model the *strategic response* of users or miners to such a policy change which we are trying to address.

2.2.2 Proposer-Builder Separation (PBS)

The Ethereum research community has proposed **Proposer-Builder Separation** (PBS) as a structural remedy: specialised *block builders* compete to construct the most profitable block, while *proposers* (validators) simply select the highest-revenue block without seeing its contents. PBS partially democratises MEV revenue by routing it through an open auction, but critics note it may entrench builder oligopolies and does not eliminate user harm from frontrunning.

2.3 Batch Auctions as a Market Design Response

Budish, Cramton, and Shim [3] provide the theoretical foundation for batch auctions in the context of traditional financial markets. Their central insight is that continuous limit order book trading creates a *latency arms race* in which any finite speed advantage allows high frequency traders to exploit stale quotes, wasting social resources on latency investment and producing a market that is efficient only in a narrow technical sense.

The authors propose **frequent batch auctions** (FBAs) as a solution. Orders are collected over a short discrete time interval (e.g., 100 ms) and cleared simultaneously at a single uniform price. This batching eliminates the value of sub-interval latency advantages because all orders received within the same batch are treated symmetrically.

Applied to AMM-based DEXs, the batch-auction mechanism would:

(i) Collect all orders submitted within a block's time window.

(ii) Compute a single clearing price that maximises trading volume subject to the AMM's invariant.

(iii) Execute all matched trades simultaneously at that price.

This eliminates within batch frontrunning and sandwich attacks, as no single transaction can be "inserted" before another within the same batch. However, it introduces new strategic considerations, like users may time their submissions relative to batch boundaries, and miners retain the ability to manipulate *which* batch a transaction is included in.

2.4 Related Protocols and Extensions

- **CoW Protocol (Coincidence of Wants)**: An off-chain solver network that attempts to match orders peer-to-peer before routing residual flow through on-chain AMMs. Solvers compete in a batch auction for the right to settle a given set of orders.
- **MEV Boost/Flashbots**: An off-protocol relay that allows searchers (MEV extractors) to submit *bundles* directly to block builders, bypassing the public mempool. This makes MEV "private" and reduces on-chain PGA style bidding wars but does not eliminate MEV itself.
- **SUAVE (Single Unifying Auction for Value Expression)**: A proposed architecture by Flashbots that decentralises the block-building role, aiming to make MEV extraction more competitive and its revenue more broadly shared.

2.5 Summary of Literature Gaps

Table 1 summarises the key works and the gaps our project addresses.

Table 1: Summary of Related Works and Identified Gaps

Work	Focus	Contribution	Gap
Daian et al. (2019) [1]	MEV measurement & PGA modeling	Quantifies MEV; formalises PGA as continuous game; grim-trigger equilibrium	No protocol-level counterfactual; no batch auctions
Alipanahloo et al. (2024) [2]	Fairness in rollup sequencing	Fairness granularity metric; randomised sequencing policy	No strategic user/miner model; no batch-auction comparison
Budish et al. (2015) [3]	HFT arms race in traditional markets	Batch auctions eliminate latency rents	Not applied to blockchain/AMM setting; no miner-strategic model
This project	Game-theoretic comparison: continuous vs. batch	Three player Stackelberg model; comparative equilibrium analysis; simulation	—

3 Formal Game Theoretic Model

We now develop a formal model of strategic interactions in a DEX ecosystem. The model is designed to be tractable while capturing the essential incentive misalignments identified in the literature.

3.1 Overview and Modelling Philosophy

We model the DEX ecosystem as a **three-player extensive-form game** with sequential moves. The game structure reflects the natural timing of events on a blockchain:

Stage 1. **User stage:** Users observe the current AMM state and submit signed transactions with chosen gas prices and trade parameters.

Stage 2. **Miner/validator stage:** The block producer observes the full mempool and chooses a transaction ordering (and possibly inserts own transactions) to maximise their objective.

Stage 3. DEX settlement stage: The protocol clears transactions according to its rules (continuous or batch), producing final balances.

This sequential structure is a **Stackelberg game**, users are *leaders* who commit to trades before the miner's ordering decision, while miners are *followers* who best respond to the observable mempool. The DEX protocol is a rule setter whose design choice (mechanism) is treated as a parameter, enabling comparative analysis.

3.2 State Space

Definition 3.1 (AMM State)

Let $\mathcal{S} = \mathbb{R}^2_{>0}$ denote the state space of a constant-product AMM holding two assets A and B. A state is a pair $(r_A, r_B) \in \mathcal{S}$ where r_A, r_B are the reserve quantities. The *spot price* of asset A in terms of B at state (r_A, r_B) is

$$P(r_A, r_B) = \frac{r_B}{r_A}.$$

After a trade of Δ_A units of A for B, the new state is $(r_A + \Delta_A,\ r_B - \Delta_B)$ where Δ_B satisfies the *constant-product invariant*:

$$(r_A + \Delta_A)(r_B - \Delta_B) = r_A \cdot r_B.$$

This gives the standard AMM output formula:

$$\Delta_B = \frac{r_B \cdot \Delta_A}{r_A + \Delta_A}.$$

3.3 Players

3.3.1 Player 1: The User Set

Definition 3.2 (Users)

Let $\mathcal{U} = \{u_1, u_2, \ldots, u_n\}$ be a finite set of *users*. Each user u_i is characterised by:

- A *desired trade*: $(\delta_i, \mathrm{dir}_i)$ where $\delta_i > 0$ is the input quantity and $\mathrm{dir}_i \in \{A \to B,\ B \to A\}$ is the trade direction.
- A *slippage tolerance*: $\tau_i \in [0, 1]$, the maximum acceptable fraction of price impact.
- A *private valuation*: $v_i > 0$, the value the user places on executing the trade at the *fair* AMM price (i.e., without adverse price impact).
- A *budget constraint* on gas: $g_i^{\max} \geq 0$.

Users are modelled as *price taking* with respect to the AMM (they take the current state as given) but *strategic* with respect to their gas bid and submission timing.

3.3.2 Player 2: The Miner/Validator

Definition 3.3 (Miner/Validator)

The *miner* M is a single strategic agent (representing the block producer) who:

- Observes the full *mempool* $\mathcal{P}$: the set of all pending signed transactions submitted by users.
- Has the unilateral power to choose any ordering $\sigma \in \mathrm{Perm}(\mathcal{P})$ of the pending transactions.
- May insert *own transactions* (e.g., frontrunning trades) at any position in the block.
- Earns revenue from (i) gas fees, (ii) OO fees from strategic ordering, and (iii) profit from own inserted trades.

Note on the miner model: In proof-of-work systems, the miner is the entity that solves the block puzzle whereas in proof-of-stake systems, this role is played by the validator/proposer. We use "miner" throughout for brevity.

3.3.3 Player 3: The DEX Protocol

Definition 3.4 (DEX Protocol)

The *DEX protocol* Π is a rule- etting entity (not a strategic player in the game-theoretic sense, but a *mechanism designer*) that specifies:

- The *clearing rule*: continuous (sequential) or batch (uniform price auction).
- The *fee structure*: trading fees $\phi \in [0, 1]$ charged on each trade.
- Optionally, a *maximum slippage* parameter $\bar{\tau}$ beyond which trades are automatically reverted.

The protocol's objective is to maximise *market efficiency* (total trading volume and price accuracy) and *user welfare* (minimising aggregate slippage).

3.4 Strategy Spaces

3.4.1 User Strategy Space

Definition 3.5 (User Strategy)

A (pure) strategy for user u_i is a tuple

$$s_i = (t_i,\ g_i,\ \delta_i,\ \tau_i^{\mathrm{set}}) \in \mathcal{S}_i^U$$

where:

- $t_i \geq 0$: submission time (relative to block start).
- $g_i \in [0, g_i^{\max}]$: gas price bid (in Gwei).
- $\delta_i > 0$: trade size (may differ from desired size as a form of demand splitting).
- $\tau_i^{\mathrm{set}} \in [0, \tau_i]$: on-chain slippage tolerance encoded in the transaction.

A *mixed strategy* for user u_i is a probability distribution $\mu_i \in \Delta(\mathcal{S}_i^U)$ over pure strategies.

Remark 3.1. In the continuous trading setting, users have an incentive to submit with high gas prices to gain priority over both other users and (defensive) bots. In the batch auction setting, gas prices no longer determine within-batch ordering, eliminating this particular strategic dimension.

3.4.2 Miner Strategy Space

Definition 3.6 (Miner Strategy)

Let $\mathcal{P} = \{x_1, \ldots, x_k\}$ denote the set of pending transactions observed by the miner. A (pure) strategy for the miner is a pair

$$s_M = (\sigma,\ \mathcal{I}) \in \mathcal{S}^M$$

where:

- $\sigma : \{1, \ldots, k\} \to \{1, \ldots, k\}$ is a permutation specifying the order in which user transactions are executed.
- $\mathcal{I} = \{y_1, \ldots, y_m\}$ is a (possibly empty) set of transactions inserted by the miner, each with a specified position in the block.

The miner's *feasibility constraint* requires the total gas used by all transactions to not exceed the block gas limit $G^{\max}$.

We distinguish three canonical miner strategies:

M1. Honest ordering: σ orders transactions by decreasing gas price; $\mathcal{I} = \emptyset$. This is the "default" miner behaviour.

M2. Frontrunning: The miner inserts own transaction(s) before a detected profitable user transaction to capture the price impact.

M3. Sandwich attack: The miner inserts a buy transaction *before* and a sell transaction *after* a large user trade to profit from the price impact it causes.

3.4.3 Sandwich Attack: Formal Specification

Definition 3.7 (Sandwich Strategy)

Given a target user trade $x_i = (\delta_i, A \to B)$, a sandwich strategy is $(\sigma, \mathcal{I})$ where $\mathcal{I} = \{y^+, y^-\}$ with:

- y^+: miner buys ϵ units of A at state (r_A, r_B), shifting the price to $P' > P$.
- User trade x_i executes at the worse price P', receiving $\Delta'_B < \Delta_B$.
- y^-: miner sells ϵ units of A back at the post-user-trade state, netting a profit.

The miner's sandwich profit is:

$$\pi_M^{\text{sandwich}} = \Delta_B^{y^-} - \Delta_B^{y^+} > 0.$$

3.5 Payoff Functions

3.5.1 User Payoff

Definition 3.8 (User Payoff)

Let $\hat{\Delta}_B^i$ be the actual output of user u_i's trade under the miner's chosen ordering, and let Δ_B^i be the "fair" output (at the state the user expected when submitting, ignoring any adversarial ordering). The payoff to user u_i is:

$$\pi_i^U = v_i \cdot \frac{\hat{\Delta}_B^i}{\Delta_B^i} - g_i \cdot c_{\text{gas}} - \mathbb{1}[\text{reverted}] \cdot g_i^{\text{used}} \cdot c_{\text{gas}}$$

where v_i is user's own valuation and c_{gas} is the ETH-per-Gwei conversion, and the indicator term charges the user for gas consumed even if the transaction reverts (as in Ethereum).

The *slippage* experienced by user u_i is:

$$\text{slip}_i = 1 - \frac{\hat{\Delta}_B^i}{\Delta_B^i} \in [0, 1].$$

3.5.2 Miner Payoff

Definition 3.9 (Miner Payoff)

The miner's total payoff from mining a block with strategy $(\sigma, \mathcal{I})$ is:

$$\pi_M = \underbrace{\sum_{i=1}^{k} g_{\sigma(i)} \cdot \text{gas}_{\sigma(i)}}_{\text{gas fees}} + \underbrace{\sum_{y_j \in \mathcal{I}} \pi_M^{(j)}}_{\text{OO / MEV}} + B_{\text{block}}$$

where B_{block} is the fixed block reward, $g_{\sigma(i)}$ is the gas price of the i-th ordered transaction, $\text{gas}_{\sigma(i)}$ is its gas consumption, and $\pi_M^{(j)}$ is the profit from the j-th inserted transaction.

In a **sandwich attack**, the miner's incremental MEV is:

$$\pi_M^{\text{MEV}} = \Delta_B^{y^-} - \Delta_B^{y^+} = \frac{r_B \epsilon}{r_A + \epsilon} - \frac{(r_B - \Delta_B^{y^+} - \hat{\Delta}_B^i)\,\epsilon}{(r_A + \epsilon + \delta_i) + \epsilon}$$

Here, we have simplified by assuming the miner uses the same ϵ for both legs of the sandwich.

3.5.3 Protocol Payoff / Social Welfare

Definition 3.10 (Social Welfare)

The *social welfare* of an outcome is:

$$W = \sum_{i=1}^{n} \pi_i^U + \pi_M - \text{deadweight loss}$$

where the deadweight loss arises from trades that are reverted due to exceeded slippage tolerances or from resources expended in the PGA bidding process that do not create value (e.g., gas burned by losing bids).

Since π_M includes MEV *extracted from* users, it is more illuminating to report **user welfare** separately:

$$W^U = \sum_{i=1}^{n} \pi_i^U.$$

3.6 Information Structure

Definition 3.11 (Information Sets)

- **Users:** Each user u_i observes the current AMM state (r_A, r_B) and the current mempool $\mathcal{P}^{-i}$ (transactions of others) with a latency $\Delta_i^U \geq 0$. Users do *not* observe the miner's ordering decision before submitting.
- **Miner:** The miner observes the *complete* mempool $\mathcal{P}$ before making its ordering decision. This is the miner's fundamental information advantage.
- **Protocol:** The protocol has no private information; it mechanically applies the clearing rule to whatever ordered list of transactions it receives.

3.7 Equilibrium Concepts

3.7.1 Nash Equilibrium

Definition 3.12 (Nash Equilibrium)

A strategy profile $(s_1^*, \ldots, s_n^*, s_M^*)$ is a *Nash equilibrium* (NE) if no player can strictly increase their payoff by unilaterally deviating.

$$\pi_i^U(s_i^*, s_{-i}^*) \geq \pi_i^U(s_i, s_{-i}^*) \quad \forall\, s_i \in \mathcal{S}_i^U,\ \forall\, i$$

$$\pi_M(s_M^*, s_{-M}^*) \geq \pi_M(s_M, s_{-M}^*) \quad \forall\, s_M \in \mathcal{S}^M.$$

3.7.2 Stackelberg Equilibrium

Definition 3.13 (Stackelberg Equilibrium)

In the *Stackelberg formulation*, users are **leaders** and the miner is the **follower**.

1. Users simultaneously choose strategies $(s_1, \ldots, s_n)$, which are observed by the miner.
2. The miner best-responds:

$$s_M^*(s_{-M}) = \arg\max_{s_M \in \mathcal{S}^M} \pi_M(s_M;\, s_1, \ldots, s_n).$$

3. A *Stackelberg equilibrium* is a profile $(s_1^*, \ldots, s_n^*,\ s_M^*)$ such that users play best-responses given the miner's best-response correspondence:

$$s_i^* \in \arg\max_{s_i} \pi_i^U(s_i, s_{-i}^*;\ s_M^*(s_1^*, \ldots, s_i, \ldots, s_n^*)).$$

Remark 3.2. The Stackelberg formulation captures the reality that users submit transactions *before* the miner selects an ordering, but *after* the miner has set up infrastructure to exploit known user patterns. Users who are aware of the miner's strategy must choose their gas bid, trade size, and slippage tolerance defensively.

3.7.3 Repeated Game and Cooperation

Drawing on the cooperative equilibrium observed by Daian et al. among PGA bots, we also consider a **repeated game** formulation where the same block is a single stage of a game repeated across many blocks.

Definition 3.14 (Repeated Game Payoff)

In the repeated game with discount factor $\beta \in (0, 1)$, player i's payoff from stage-game payoff sequence $(\pi_i^t)_{t=0}^{\infty}$ is:

$$\Pi_i = (1 - \beta) \sum_{t=0}^{\infty} \beta^t \pi_i^t.$$

The Folk Theorem guarantees that any feasible and individually rational payoff profile can be sustained in a subgame-perfect equilibrium of the infinitely repeated game, provided β is sufficiently close to 1. This provides theoretical grounding for the cooperative equilibria observed empirically.

3.8 Priority Gas Auction Sub-Game

When multiple users (or bots) compete for the same block position, their interaction is modelled as a PGA sub-game, formalised following Daian et al.:

Definition 3.15 (PGA Sub-Game)

A *Priority Gas Auction* among n competitors for a single block position is a sequential game with:

- **Continuous time:** Bids can be submitted at any time $t \in [0, T)$ where $T \sim \mathrm{Exp}(\lambda)$ is the random block-mining time.
- **All-pay:** The losing player pays $\ell(\$b_{\text{last}})$ where $\$b_{\text{last}}$ is their final gas bid and $\ell(\cdot)$ is a loss function ($\ell(b) < b$, partial all-pay).
- **Minimum raise constraint:** Any new bid must exceed the player's previous bid by at least a factor $(1 + \iota)$, where $\iota = 0.125$ under Parity client defaults.
- **Imperfect information:** Player i observes competitor bids with latency $\Delta_i \geq 0$.

Theorem 3.1 (Cooperative Nash Equilibrium in PGAs [1])**.** *For parameters consistent with Ethereum PGAs (exponential block time with $\lambda^{-1} = 15\,s$, minimum raise $\iota = 0.125$, mean inter-bid interval $\delta = 0.4\,s$, latency $\Delta_i = 0.1\,s$), there exists a* grim trigger Nash equilibrium *in which two competing players alternate bids at minimum-raise increments, and any deviation is met with an immediate bid to the maximum profitable level.*

3.9 The Batch Auction Extension

Definition 3.16 (Batch Auction Mechanism)

Under a *batch auction* clearing rule, the DEX protocol:

(i) Collects all transactions submitted within block interval $[0, T)$.

(ii) Determines a *uniform clearing price* P^* that maximises the volume of matched buy and sell orders subject to the AMM invariant.

(iii) Executes all matched trades simultaneously at P^*.

Formally, P^* solves:

$$P^* = \arg\max_P \min\left(\sum_{i:\,\mathrm{dir}_i = A \to B,\, v_i \geq P} \delta_i, \sum_{j:\,\mathrm{dir}_j = B \to A,\, v_j \leq P} \delta_j \right).$$

Proposition 3.1 (Elimination of Within-Batch Frontrunning)**.** *Under the batch auction mechanism, for any two transactions x_i, x_j submitted within the same batch, the miner's choice of ordering σ over $\{x_i, x_j\}$ does not affect their individual execution prices. Consequently, no sandwich attack is profitable within a batch.*

Proof sketch. Since all trades within a batch execute at the uniform clearing price P^*, the miner has no mechanism to insert a transaction between x_i and x_j that would shift the price at which either trade executes. Any miner-inserted transaction y within the batch is also subject to the same clearing price P^*, eliminating the asymmetric price impact that makes sandwiching profitable. □

4 Comparative Analysis: Continuous vs. Batch Clearing

4.1 Equilibrium Outcomes Under Continuous Trading

Under continuous AMM trading, the Stackelberg equilibrium is characterised by the following dynamics:

(i) **Miner best response:** For any user transaction with trade size δ_i large enough that the slippage $\text{slip}_i > 0$, the miner's dominant strategy (given full mempool visibility) is to sandwich the trade if:

$$\pi_M^{\text{sandwich}}(\delta_i) > \text{cost of inserted gas.}$$

(ii) **User best response:** Anticipating sandwiching, rational users set low slippage tolerances τ_i^{set} and split large trades into smaller chunks—both of which reduce sandwich profitability but also reduce trading efficiency.

(iii) **Gas price war:** Multiple users competing for priority face a PGA sub-game (Definition 3.15) over gas prices, driving bids up and wasting value in the all-pay component.

4.2 Equilibrium Outcomes Under Batch Auctions

Under batch-auction clearing:

(i) **Elimination of within-batch MEV:** By Proposition 3.1, sandwich attacks are infeasible within a batch. Frontrunning across batches (i.e., the miner choosing *which* batch to include a transaction in) remains possible but is coarser and less profitable.

(ii) **User strategy simplification:** Users need not set defensive slippage limits to protect against within batch ordering; their only strategic variable becomes submission timing relative to batch windows.

(iii) **Residual miner MEV:** The miner retains the ability to delay or accelerate a transaction's batch membership. However, the value of this is bounded by the price movement between consecutive batches, which is smaller than within-transaction slippage under continuous trading for reasonable batch durations.

4.3 Welfare Comparison

Table 2 summarises the theoretical welfare predictions.

Table 2: Predicted Welfare Outcomes: Continuous vs. Batch Clearing

Metric	**Continuous AMM**	**Batch Auction**
User slippage (avg.)	High (sandwich attacks)	Low (uniform price)
Miner MEV revenue	High	Low/Moderate
Gas fee waste (PGAs)	High	Low (no priority race)
Trade execution latency	Low (immediate)	Higher (batch window)
Price discovery efficiency	Continuous	Batch interval resolution
Frontrunning vulnerability	High	Low (within batch)
Consensus security risk	High (OO fees)	Reduced

5 Discussion and Next Steps

5.1 Key Theoretical Insights

The formal model developed in §3 yields several important theoretical insights before any simulation is run:

I1. Miner dominance under continuous trading: The miner's last-mover advantage in the Stackelberg game, combined with full mempool visibility, gives them almost complete control over user execution prices. No user-level defense (slippage limits, trade splitting) can fully neutralise this.

I2. Batch auctions shift the Stackelberg leader: Under batch clearing, the DEX mechanism becomes the effective "first mover" by committing to a clearing rule that constrains the miner's strategy space. This is a qualitative shift in the game's structure.

I3. Cooperative MEV among bots: Theorem 3.1 suggests that competing arbitrage bots may settle into cooperative equilibria that benefit *miners* (through predictable fee revenue) at the expense of *users* (through sustained MEV extraction), even without explicit collusion.

I4. Batch auctions are not MEV-free: Batch clearing reduces but does not eliminate MEV. Between-batch manipulation and solver-collusion risks (in protocols like CoW) require additional mechanism design.

5.2 Plan for Phase 2

In the second phase we plan to:

- Implement the transaction generator and continuous AMM simulation in Python.
- Implement the batch-auction clearing mechanism.
- Simulate miner reordering strategies (honest, frontrunning, sandwich) against defensive and naive user strategies.
- Compute empirical payoffs and compare against theoretical equilibrium predictions from §3.
- Conduct preliminary comparative welfare analysis.

6 Conclusion

This report has laid the theoretical groundwork for our project. We reviewed the foundational literature on MEV, Priority Gas Auctions, and batch-auction alternatives, identifying a clear gap in the game-theoretic comparative analysis of these mechanisms. We then developed a formal three player Stackelberg model that precisely defines the players, strategy spaces, payoff functions, and equilibrium concepts appropriate for our setting.

The model reveals that continuous AMM trading creates a structurally advantaged miner who can extract substantial value from users, while batch-auction clearing fundamentally alters the game's information structure to the users' benefit. These theoretical predictions will be validated and quantified through simulation in Phase 2.

References

[1] Philip Daian, Steven Goldfeder, Tyler Kell, Yunqi Li, Xueyuan Zhao, Iddo Bentov, Lorenz Breidenbach, and Ari Juels. Flash boys 2.0: Frontrunning, transaction reordering, and consensus instability in decentralised exchanges. *arXiv preprint arXiv:1904.05234*, 2019.

[2] Zeinab Alipanahloo, Kaiwen Zhang, and Emmanuel Awosika. Quantifying fairness granularity as a fair ordering policy towards MEV mitigation for rollups. In *2024 IEEE International Conference on Blockchain (Blockchain)*, pages 209–218. IEEE, 2024.

[3] Eric Budish, Peter Cramton, and John Shim. The high-frequency trading arms race: Frequent batch auctions as a market design response. *The Quarterly Journal of Economics*, 130(4):1547–1621, 2015.

[4] Miles Carlsten, Harry Kalodner, S. Matthew Weinberg, and Arvind Narayanan. On the instability of bitcoin without the block reward. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, pages 154–167, 2016.

[5] Martin Shubik. The dollar auction game: A paradox in noncooperative behavior and escalation. *Journal of Conflict Resolution*, 15(1):109–111, 1971.